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Turtle Games: Predicting Customer Loyalty Outcomes

Date: 24 April 2026

ABSTRACT

This report analyses customer data for Turtle Games, a global manufacturer and retailer of games products. It investigates how customers accumulate loyalty points, the factors associated with loyalty, the segments within the customer base, and the suitability of predictive models for retention planning.

The analysis combines exploratory data analysis, regression, clustering, decision-tree modelling, and natural language processing. Together, these methods provide evidence on customer behaviour, customer sentiment, and opportunities for targeted marketing.

Spending score is a stronger predictor of loyalty points than demographic variables, and five distinct customer segments were identified. A multiple linear regression model explained 84% of the variation in loyalty points ($R^2 = 0.8399$), although its diagnostic results mean that predictions should be interpreted with caution. Overall sentiment was predominantly positive, but a small number of strongly negative reviews warrant investigation.

BACKGROUND AND BUSINESS CONTEXT

Turtle Games needs a clearer understanding of the behaviour that drives customer loyalty. The objective of this analysis is to identify patterns and provide evidence-based recommendations that can inform marketing and sales decisions, including resource allocation, targeted campaigns, customer retention, and opportunities to increase engagement among under-served segments.

- What are the key drivers of customer loyalty points?
- Are there meaningful segments within the customer base?
- What is the overall sentiment expressed in customer feedback?
- Which model could support the prediction of loyalty points for retention strategies?

ANALYTICAL APPROACH

This section summarises the data, tools, and analytical procedures used in the project.

Data: The analysis used `turtle\_reviews.csv`, which contains numerical customer attributes and text-based review data, together with supporting metadata.

Python and Jupyter Notebook

Data cleaning and preparation

Data preparation was completed primarily in a Python Jupyter Notebook. The process included the following steps:

- Removed the language and platform columns because they were not required for the analysis.
- Renamed the remuneration and spending-score fields to clear, analysis-ready names.
Saved the prepared dataset as `clean\_reviews.csv`.

Python libraries and packages

The following libraries and packages were used:

- NumPy and pandas
- Matplotlib and Seaborn
- statsmodels
- scikit-learn (including DecisionTreeRegressor)
- SciPy
- TextBlob, NLTK, and WordCloud
- K-means, elbow, and silhouette-score tools

These tools were used to create scatter plots, pair plots, boxplots, bar charts, histograms, cluster plots, and model diagnostics.

Data loading

The dataset was loaded into the notebook using pandas and the `pd.read_csv()` function.

Decision-tree modelling

The initial decision tree was large and difficult to interpret. Cost-complexity pruning was applied, and text variables (review and summary) were excluded to reduce complexity. Despite these steps, the final tree remained complex and should be used cautiously for business decision-making.

R and RStudio

The R analysis used the previously prepared `clean_reviews.csv` dataset; no substantial additional data cleaning was required.

R packages

The following packages were used in R:

- `ggplot2`
- `dplyr`
- `tidyverse`

Data loading and modelling

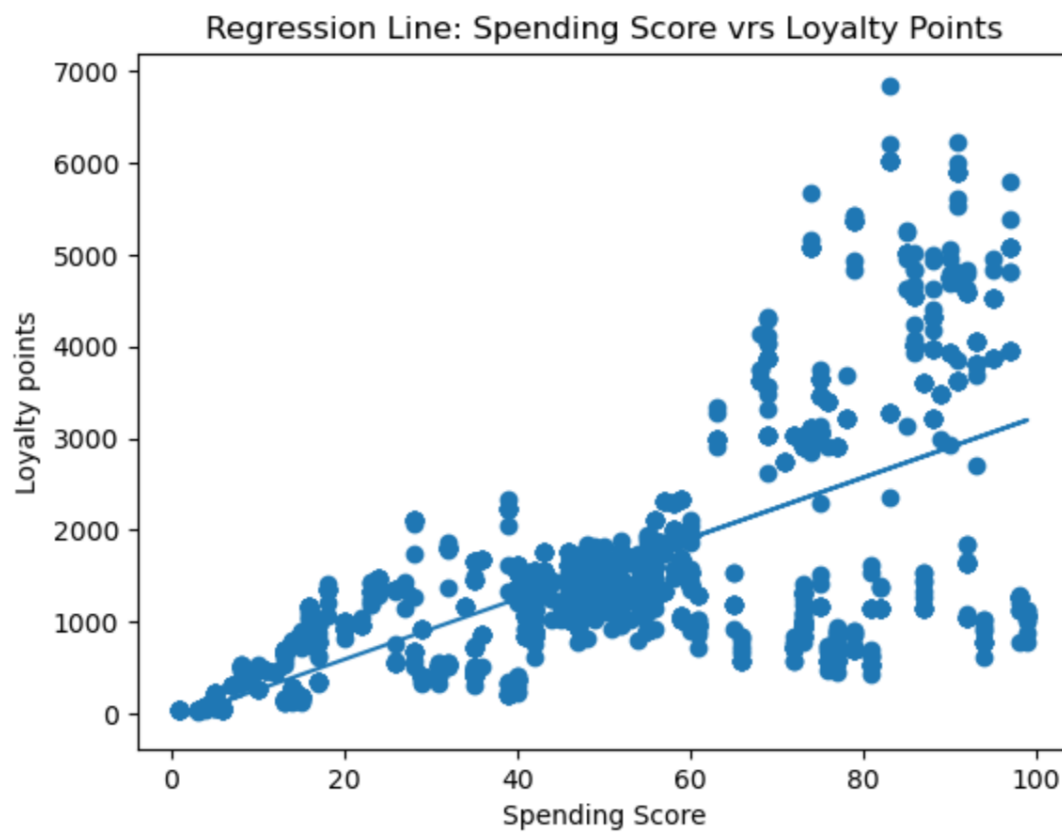
The cleaned dataset was imported in R using `read.csv()`. R was used for exploratory visualisation, multiple linear regression, diagnostic plots, and a customer-prediction scenario.

VISUALISATIONS AND INSIGHTS

This section explains the purpose of each visualisation and summarises the insight it provides.

Figure 1. Spending Score versus Loyalty Points

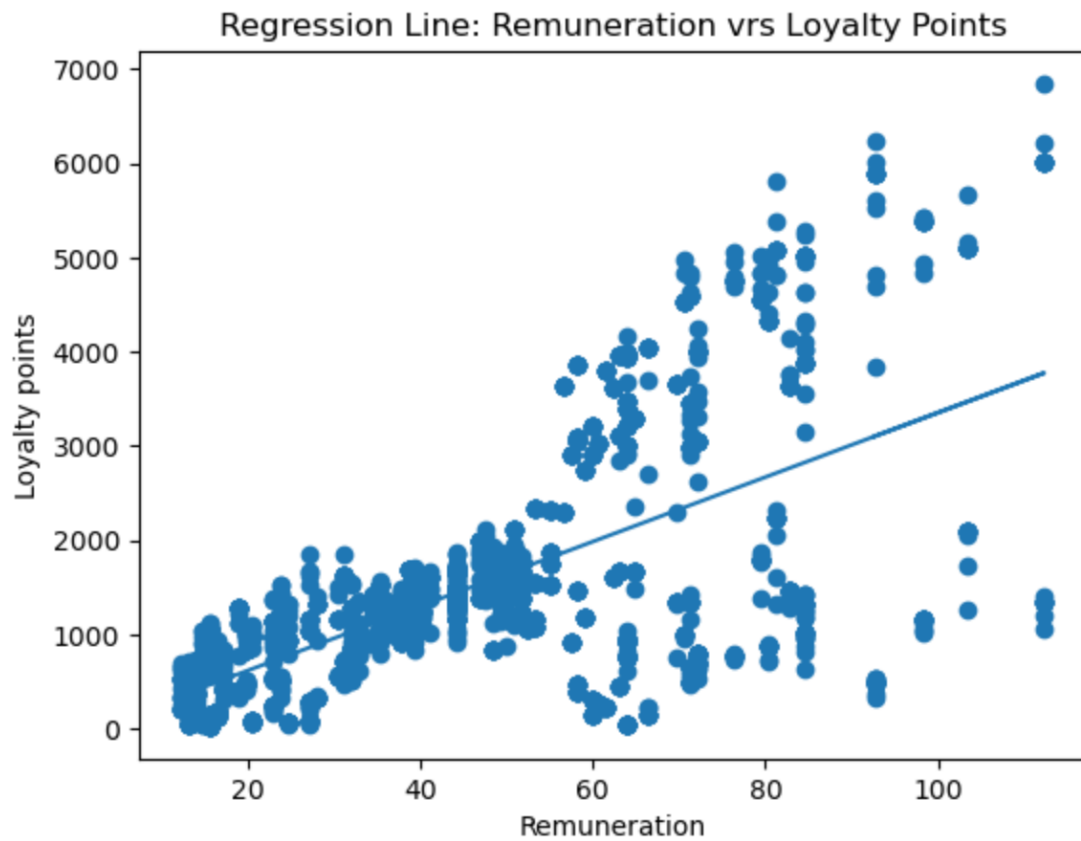
This scatter plot assesses the relationship between spending score and loyalty points.



- The plot shows a positive relationship between spending score and loyalty points.
- The model has an R^2 of 0.452, indicating that spending score explains 45.2% of the variation in loyalty points. This represents moderate explanatory power (see Appendix 1).

Figure 2. Remuneration versus Loyalty Points

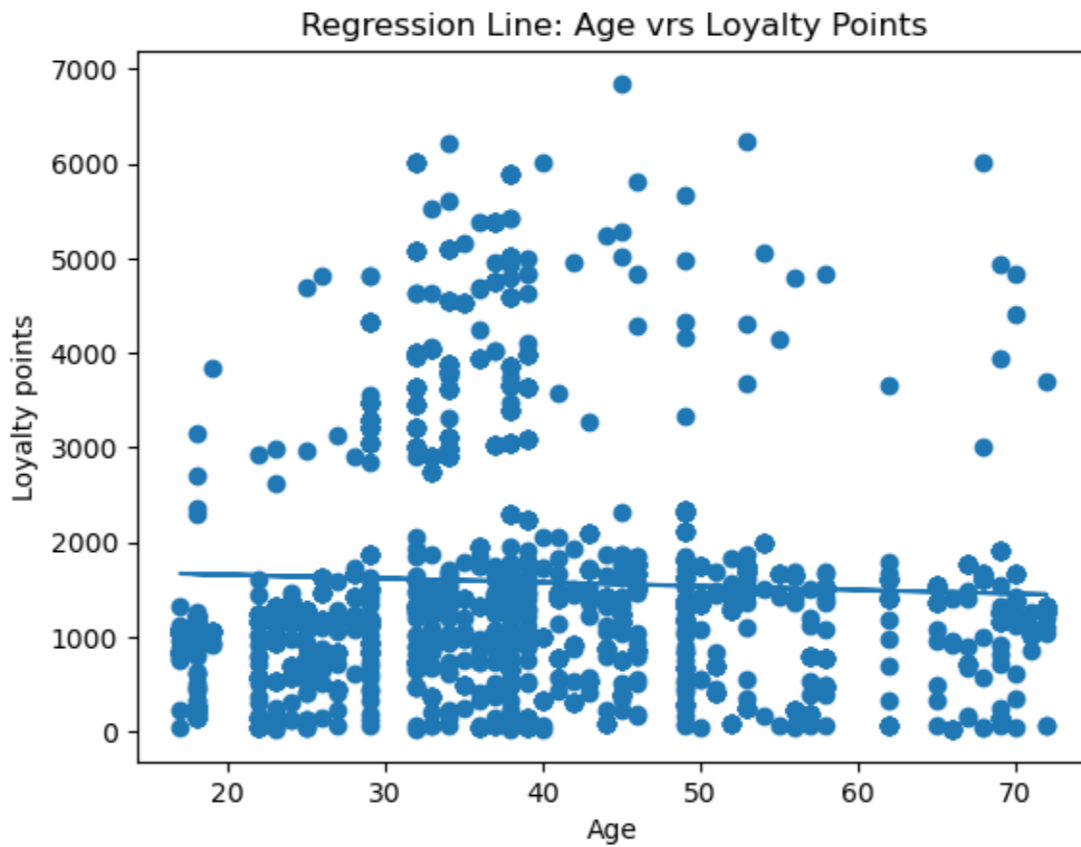
This scatter plot assesses whether remuneration is associated with loyalty points.



- The relationship between remuneration and loyalty points is positive.
- The model has an R^2 of 0.38, meaning remuneration explains 38% of the variation in loyalty points. This represents moderate explanatory power (see Appendix 2).

Figure 3. Age versus Loyalty Points

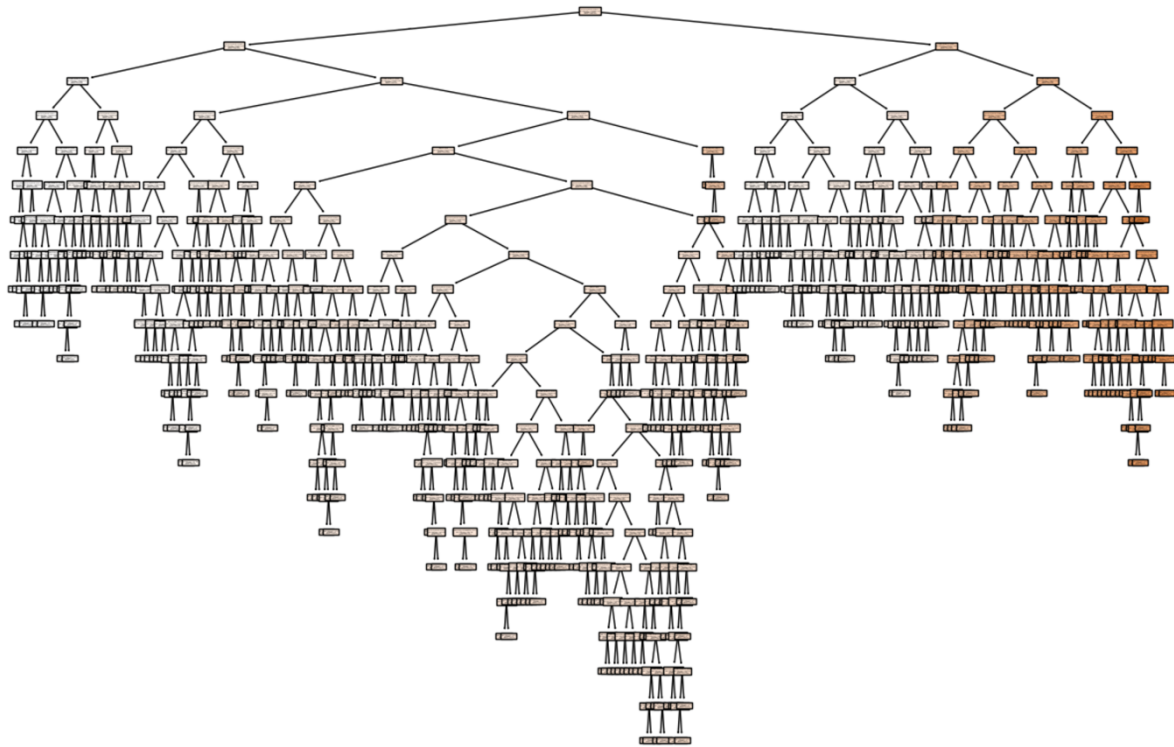
This scatter plot assesses whether age is associated with loyalty points.



- The plot shows no meaningful relationship between age and loyalty points.
- $R^2 = 0.002$, meaning age explains only 0.2% of the variation in loyalty points. This is an extremely weak relationship (see Appendix 3).

Figure 4. Decision-Tree Regression Model

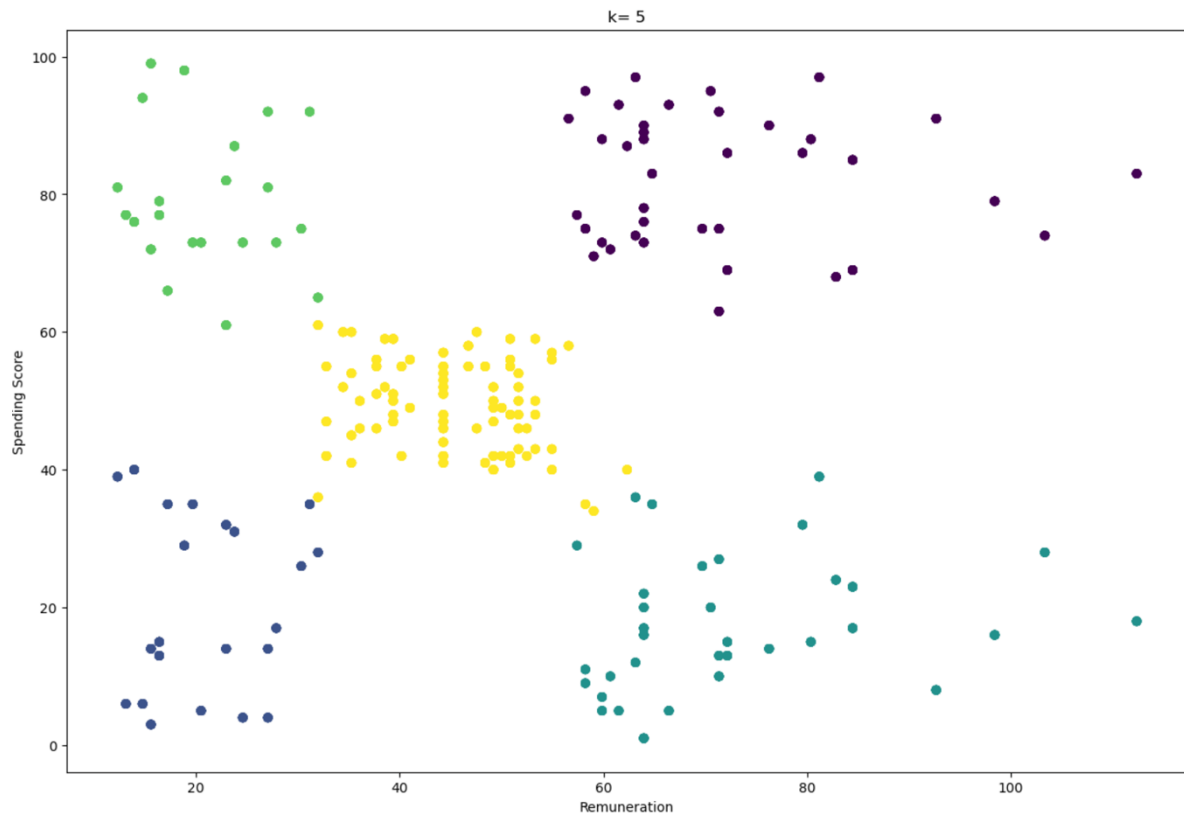
This decision-tree model estimates loyalty points from spending score, remuneration, and age.



- The model achieved $R^2 = 0.9937$ (99.37%). Although this appears excellent, the result is suspiciously high and may indicate overfitting. The model may therefore not generalise well to unseen data (see Appendix 4).

Figure 5. Customer Segments

This cluster plot identifies customer segments using remuneration and spending score.

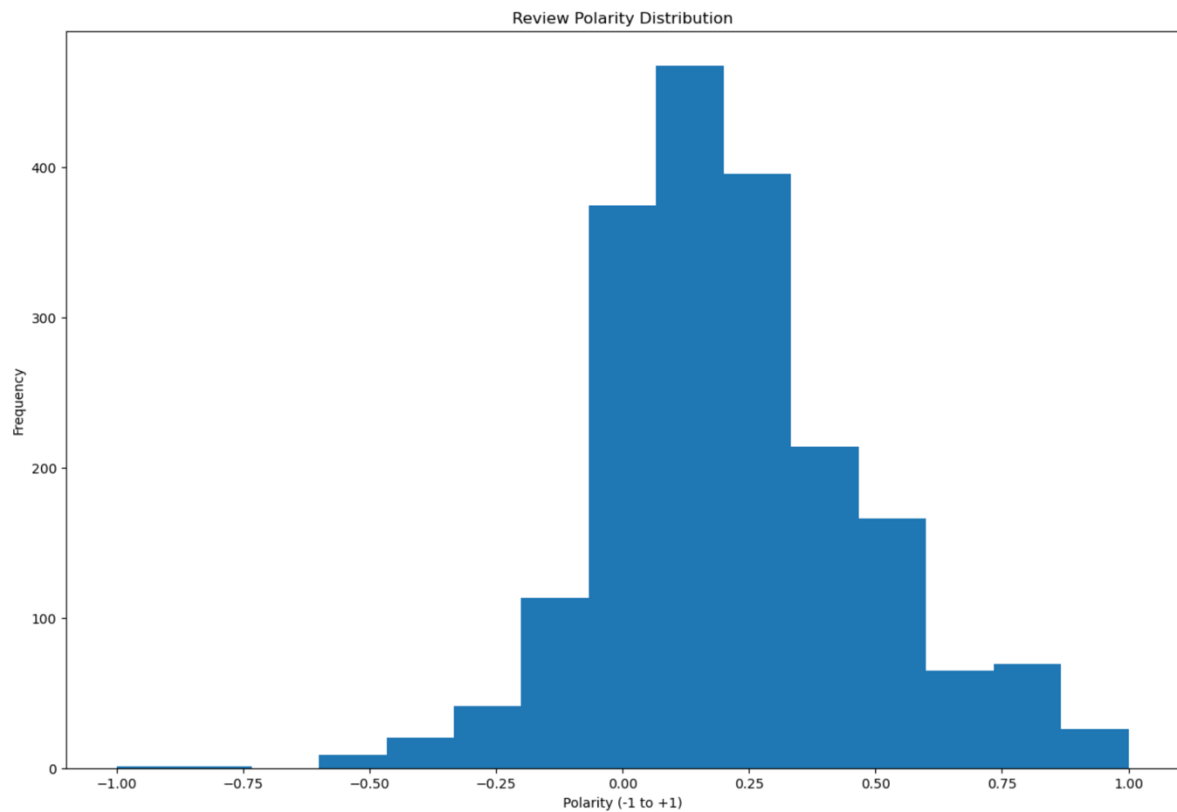


K-means clustering identified five distinct customer groups (see Appendix 5):

- High income, high spend: premium customers.
- Low income, high spend: value seekers.
- Low income, low spend: lower-value customers.
- High income, low spend: an under-utilised segment with growth potential.
- Middle segment: core customers who should be prioritised for retention.

Figure 6. Review Sentiment Polarity

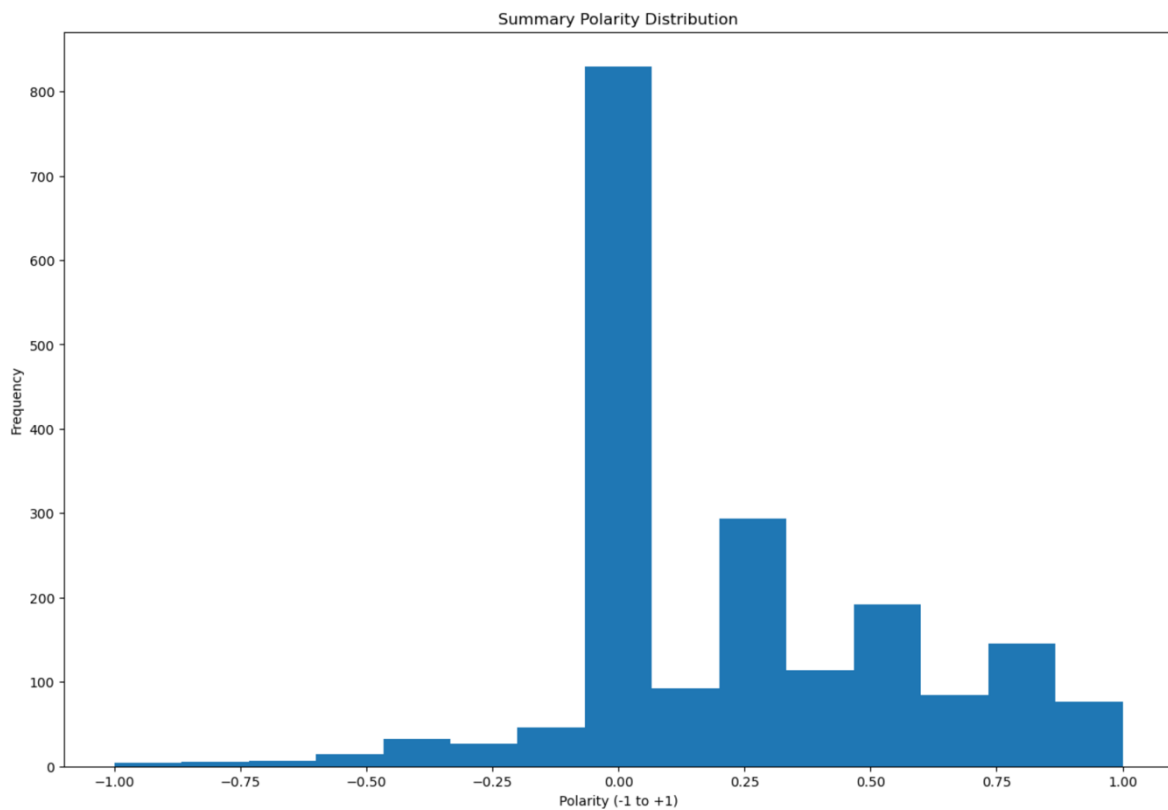
This histogram shows the sentiment expressed in customer reviews.



- Review polarity is skewed towards positive values, with most scores between 0 and 0.3. This suggests that customers are generally mildly positive rather than strongly polarised.
- There are very few strongly negative review scores (below -0.5).
- See Appendix 6 for further detail.

Figure 7. Summary Sentiment Polarity

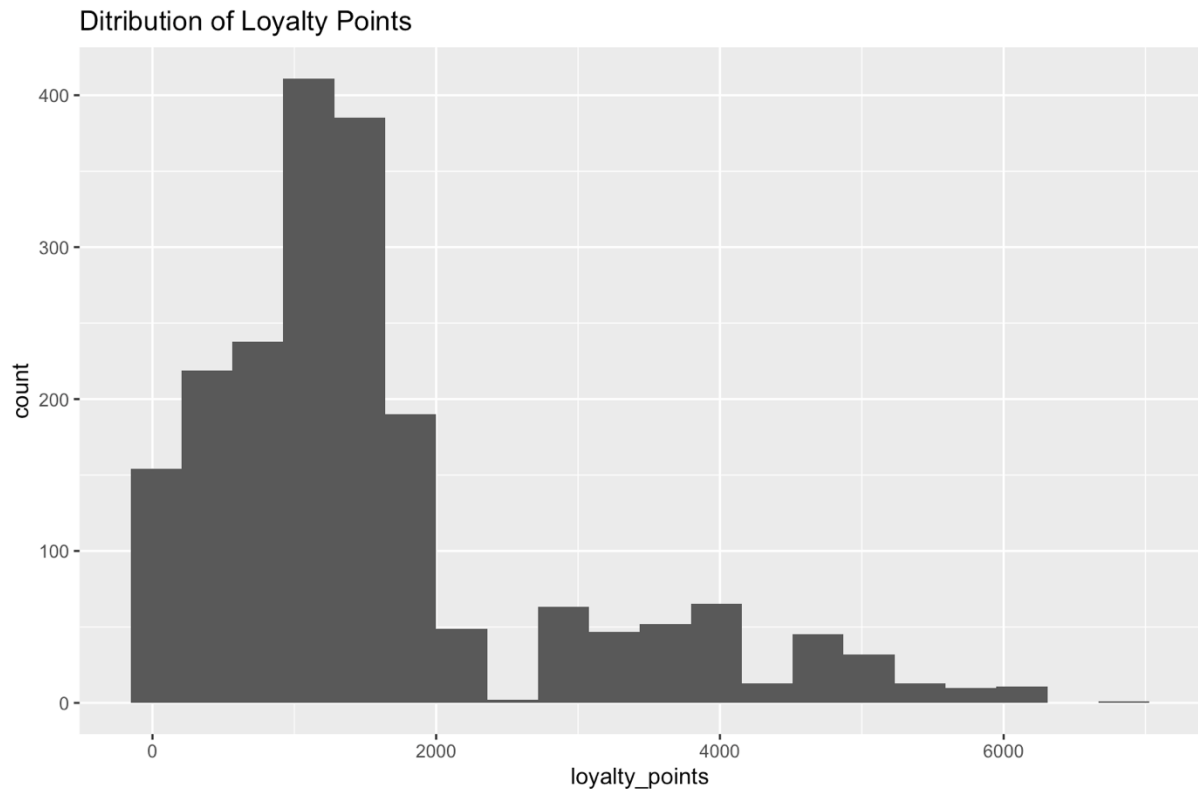
This histogram shows the sentiment expressed in customer review summaries.



- Summary sentiment is generally positive.
- Scores are concentrated around 0 (neutral) and extend into positive values between approximately 0.2 and 0.8.
- There are relatively few negative summaries.
- The summary distribution is more compressed than the review distribution, suggesting that customers use less expressive language in summaries.

Figure 8. Distribution of Loyalty Points

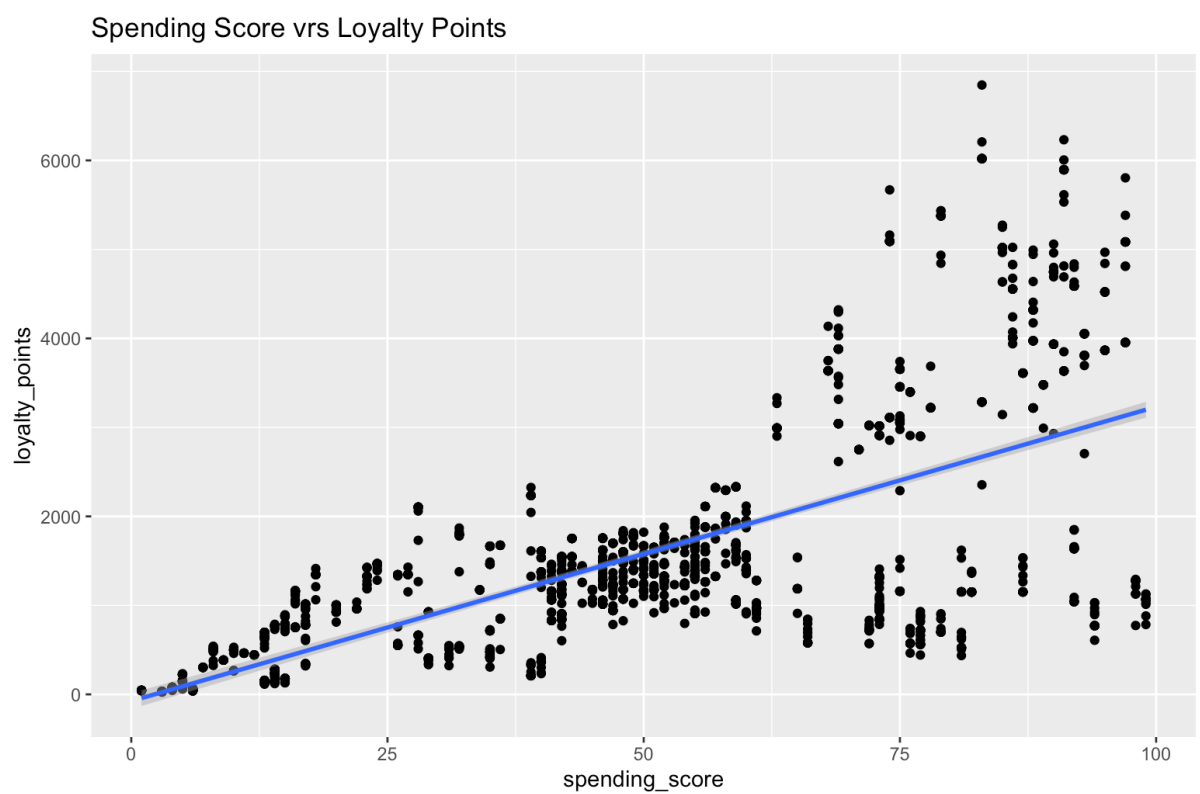
This R histogram shows the distribution of loyalty points across the customer base.



- The distribution of loyalty points is right-skewed.
- Most customers have approximately 500 to 2,000 loyalty points.
- A long right-hand tail shows that a smaller number of customers have between 3,000 and 6,500 points.
- Most customers earn moderate loyalty points and may have scope for increased engagement.
- A small group accumulates very high loyalty points and represents a high-value customer segment.

Figure 9. Spending Score versus Loyalty Points (R)

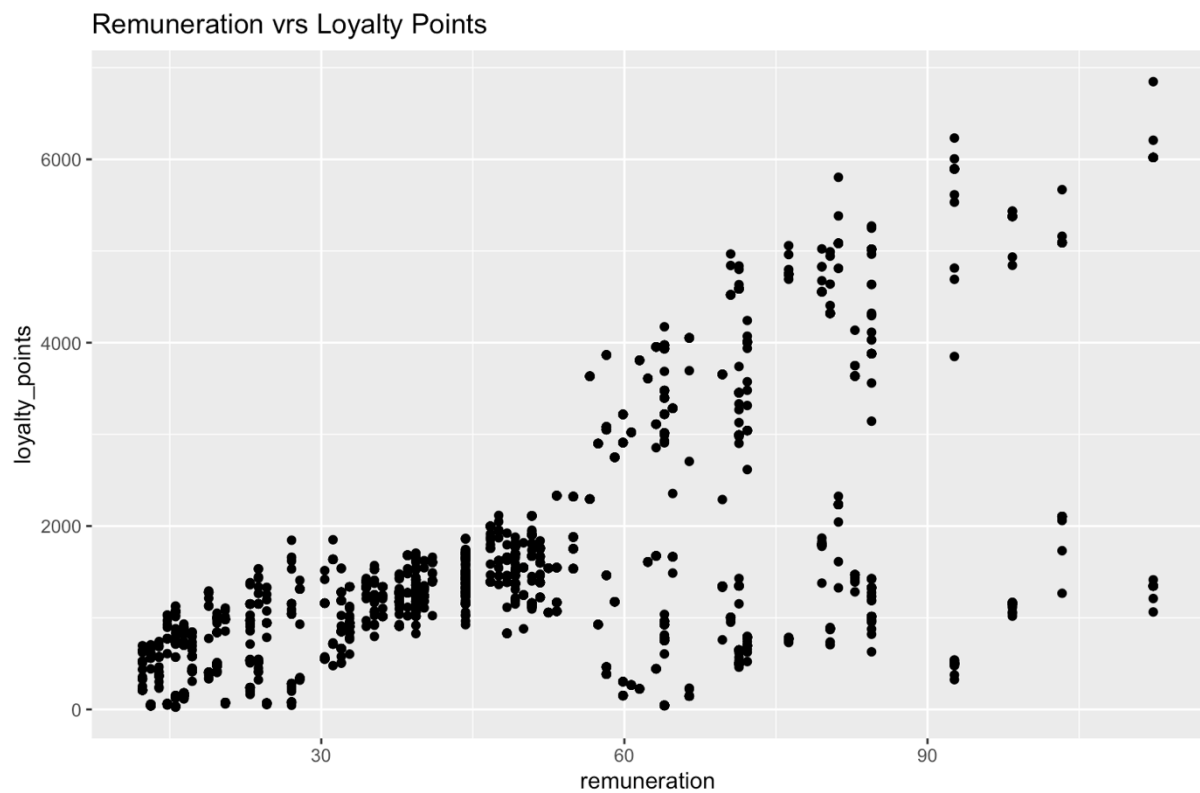
This R scatter plot examines the relationship between spending score and loyalty points.



- The relationship is positive: higher spending scores are generally associated with more loyalty points.
- The spread of points indicates that spending score alone does not explain all differences in loyalty behaviour.

Figure 10. Remuneration versus Loyalty Points (R)

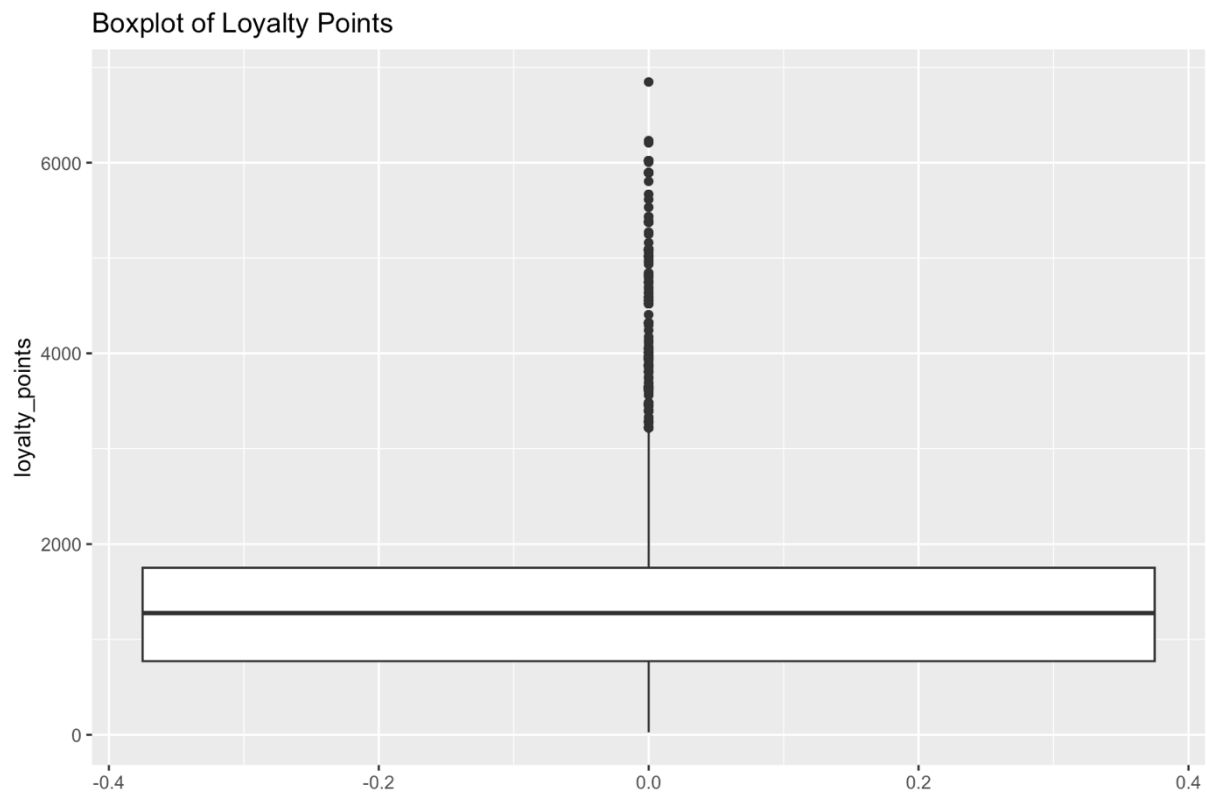
This R scatter plot examines the relationship between remuneration and loyalty points.



- The relationship is moderately positive, although it becomes more variable at middle and higher remuneration levels.
- The wide spread indicates that higher remuneration does not always lead to higher loyalty points.
- The plot suggests three broad groups:
 - Lower-income customers (20-40) with generally lower loyalty points.
 - Middle-income customers (40-70) with mixed, moderate-to-high loyalty behaviour.
 - Higher-income customers (70-90) with a wide range of loyalty outcomes, including notable outliers.

Figure 11. Boxplot of Loyalty Points

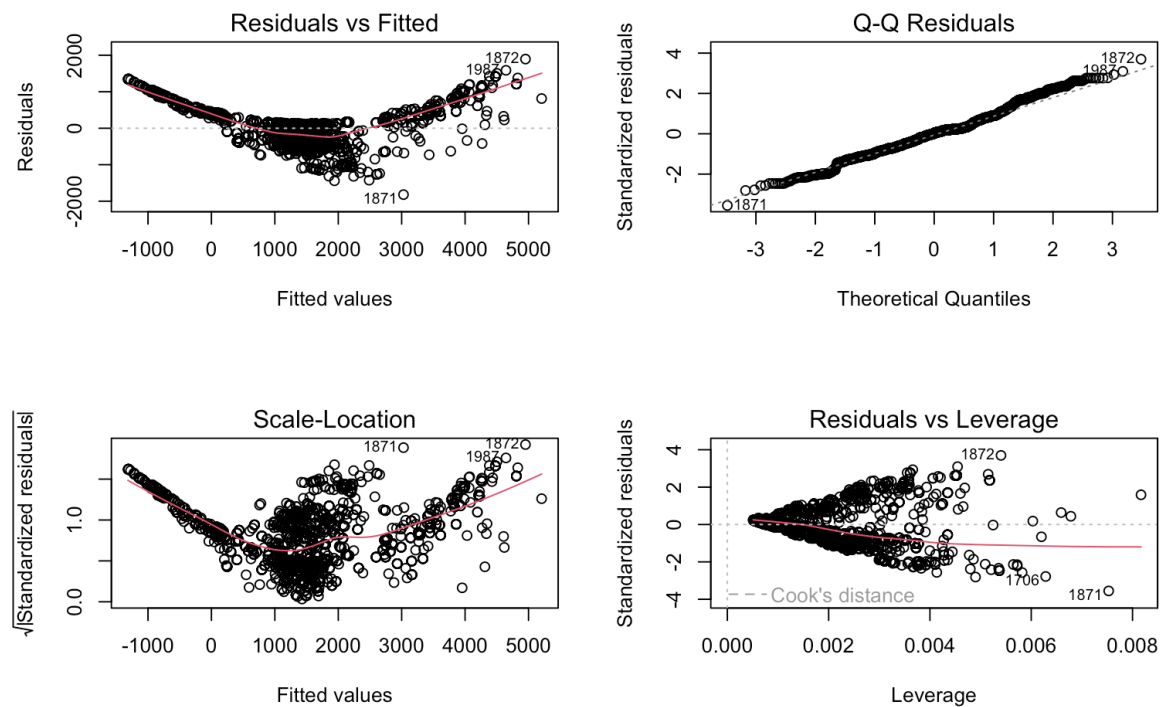
This R boxplot shows the variability and outliers in loyalty points.



- The median is approximately 1,200 to 1,400 points.
- The interquartile range shows that most customers have approximately 700 to 1,800 points.
- There are many high-value outliers above 3,000 points.
- There are fewer low-end outliers than high-end outliers.

Figure 12. Multiple Linear Regression Diagnostics

These diagnostic plots assess the reliability of the multiple linear regression model.



■ Residuals versus fitted values

- The residual pattern is not random and shows a U-shape.
- This suggests non-linearity.
- The predictors do not fully explain loyalty points through a straight-line relationship.

■ Normal Q-Q plot

- Most points follow the line, but the tails deviate.
- Residuals are not perfectly normally distributed.
- Outliers are present and may reduce the reliability of statistical tests.

■ Scale-location plot

- The residual spread increases rather than remaining constant.
- This indicates heteroscedasticity: prediction error increases as loyalty points increase.
- The model is therefore less stable for high-value customers.

■ Residuals versus leverage

- The plot identifies potentially influential observations.
- Several customers have relatively high leverage.

- A small number of high-loyalty customers may disproportionately influence the regression results.

PATTERNS, PREDICTIONS, AND RECOMMENDATIONS

Exploratory data analysis

Loyalty points are positively skewed. Spending score and remuneration are both positively associated with loyalty, but spending score has the stronger relationship. This suggests that customer behaviour is more important than demographic characteristics such as age.

Decision-tree model

The decision tree highlights spending score and remuneration as the most important predictors. However, its complexity and very high R^2 make it difficult to interpret and raise concerns about overfitting.

Customer segmentation

K-means clustering identified five segments with different spending and engagement patterns. These segments can support targeted marketing and retention activity.

Sentiment analysis

Customer feedback is predominantly positive, indicating general satisfaction with Turtle Games products. However, strongly negative reviews should be monitored and investigated to identify service or product issues.

Multiple linear regression

The multiple linear regression model predicts loyalty points from age, remuneration, and spending score, with $R^2 = 0.8399$. Although it has strong explanatory power, diagnostic plots show non-linearity, unequal variance, and influential outliers. It should be used for broad insight rather than precise high-value-customer predictions.

Prediction scenario

For a customer aged 35 with remuneration of 60 and a spending score of 80, the model predicts approximately 2,959 loyalty points. This customer is above the typical range and presents a useful opportunity for targeted retention activity; however, the prediction should be treated cautiously because the model is less stable at higher loyalty values.

APPENDIX

1. Spending Score versus Loyalty Points

The plot shows a positive relationship between spending score and loyalty points.

- The coefficient ($\beta = 33.06$) indicates that a one-unit increase in spending score is associated with an increase of approximately 33 loyalty points.
- The model is statistically significant ($F = 1648$, $p < 0.001$).

- $R^2 = 0.452$, so spending score explains 45.2% of the variation in loyalty points.

2. Remuneration versus Loyalty Points

There is a positive and statistically significant relationship between remuneration and loyalty points.

- The coefficient ($\beta = 34$) indicates that a one-unit increase in remuneration is associated with an average increase of approximately 34 loyalty points.
- $R^2 = 0.38$, so remuneration explains 38% of the variation in loyalty points.

3. Age versus Loyalty Points

The plot shows no meaningful relationship between age and loyalty points.

- $R^2 = 0.002$, so age explains only 0.2% of the variation in loyalty points.
- Adjusted $R^2 = 0.001$, indicating almost no explanatory power.
- The F-statistic p-value is 0.0577, which is above the 5% significance threshold. There is therefore weak evidence of a relationship between age and loyalty points.

4. Decision-Tree Regression Model

- The decision tree achieved $R^2 = 0.9937$ (99.37%). This very high value is likely to reflect overfitting, particularly because the tree is large and detailed. The model may not generalise well to unseen data and is difficult to use directly for business decisions.
- The top splits indicate that spending score and remuneration are key factors separating high- and low-loyalty customers.

5. Customer Segmentation

- High income, high spend: premium customers. Recommended action: VIP programmes, exclusive offers, and enhanced loyalty rewards.
- Low income, high spend: value seekers. Recommended action: discounts, promotions, and bundle offers.
- Low income, low spend: lower-value customers. Recommended action: cost-efficient, low-touch engagement.
- High income, low spend: under-utilised customers with growth potential. Recommended action: targeted campaigns and incentives to increase spend.
- Core customers: prioritise retention strategies and relevant loyalty communications.

6. Review Polarity Distribution

- Review polarity is skewed towards positive values, with most scores between 0 and 0.3.
- Most observations fall within the positive range.
- The highest concentration of scores is between 0 and approximately 0.3.
- Some reviews extend to strongly positive values between 0.5 and 1.0.
- There are very few strongly negative review scores (below -0.5).